

# KISSAN: Procure Smart Mandi

Digital Slot Booking, Vector QR Gate Passes, Live Yard Traffic Control & Instant Form 'J' MSP Payouts

**Problem in One Sentence:** Indian farmers face chaotic 12–24 hour queues outside APMC Mandis with middlemen exploitation and paperwork delays. **KISSAN** replaces manual congestion with a streamlined, digital end-to-end ecosystem: Slot Booking -> Dynamic QR Pass -> Camera Scan Gate In -> Live Yard Queue -> Weighbridge Scale -> Form 'J' DBT Receipt.

### The Traditional Mandi Problem

- **Massive Yard Congestion:** 12–24 hour tractor traffic jams during peak harvest days.
- **Middlemen & Underweighing:** Lack of digital weighbridge transparency.
- **Paperwork Friction:** Manual physical Form 'J' slips get lost or delayed.
- **No Pre-Arrival Scheduling:** Thousands of farmers arrive simultaneously.

### The KISSAN Smart Solution

- **Pre-Scheduled Arrival Slots:** 3 balanced daily windows (Morning, Noon, Afternoon).
- **Zero-Delay Vector QR Passes:** Instant, tamper-proof entry passes.
- **Mobile Camera Gate Scanner:** High-speed check-in without costly hardware.
- **Automated Form 'J' & MSP:** Direct calculations with DBT bank audit trail.

## 1. Complete Technology Stack & Architecture

| LAYER               | TECHNOLOGY USED                 | CORE PURPOSE & ADVANTAGE                                                                                                  |
|---------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Frontend UI         | HTML5 + Modern CSS3             | Zero-dependency, ultra-lightweight, 100% responsive, renders smoothly on low-bandwidth rural 2G/3G mobile devices.        |
| Client Logic        | Vanilla JavaScript (ES6+)       | No heavy framework overhead (No React/Angular/Vue), instantaneous browser execution, zero bundle size issues.             |
| Backend & Database  | Supabase (PostgreSQL 15)        | Scalable relational database with Row Level Security (RLS), real-time subscriptions, and auto-generated REST APIs.        |
| Authentication      | Supabase Auth (RBAC)            | Role-Based Access Control cryptographically distinguishing FARMER vs OFFICER/ADMIN workflows.                             |
| QR Code Engine      | Custom SVG Engine (KissanQR)    | <b>100% Offline-Ready</b> vector SVG QR generation in <b>0ms</b> ; handles regional UTF-8 crop names without CDN latency. |
| Computer Vision     | HTML5-QRCode Camera Scanner     | Turns any smartphone/tablet camera or gate terminal webcam into an instant high-speed QR pass validator.                  |
| State & Cache       | KissanDB (Hybrid Cache)         | Dual-layer offline-first architecture; persists bookings locally and synchronizes bidirectionally with Supabase.          |
| Export & Compliance | CSS Print Engine (@media print) | Generates standard government-compliant A4 and thermal printable Gate Passes and official Form 'J' Sale Slips.            |

## 2. How the System Works (End-to-End Workflow)

- 1

**Farmer Profile & Authentication**  
Farmer logs in via Supabase Auth. Profile details (Name, Mobile, Landholding, District) are safely retrieved via RLS.
- 2

**Smart Slot Booking & Instant QR Pass Generation**  
Farmer selects Mandi hub, crop (e.g. *Wheat @ ₹2,275/Qt*), quantity, and preferred arrival window. The system assigns a unique Token (KMN-046) and instantly renders a tamper-proof SVG QR Code.
- 3

**Mandi Gate 1: Live Mobile Camera Scanning & Admission**  
The gate security officer points their smartphone/tablet camera at the farmer's pass. The system decodes the payload, verifies booking authenticity, and admits the tractor into the yard with 1 click.
- 4

**Real-Time Digital Yard Queue Board**  
Vehicles move systematically across 3 stages: *Gate Entry -> Yard Transit -> Weighbridge Scale*. Farmers can view live "Now Serving Token" numbers from their phones.
- 5

**Digital Weighbridge & Moisture Quality Grading**  
Mandi operators record Gross Weight, Tare Weight, and Moisture % (FAQ standards). Net Produce Weight and total MSP value are calculated automatically.
- 6

**Digital Form 'J' Mandi Receipt & DBT Settlement**  
An official Government Form 'J' digital certificate is issued containing the transaction ID, verified QR stamp, and DBT bank reference for immediate payout verification.

## 3. Key Technical Innovations & USPs

- Zero Hardware Barrier**  
No dedicated barcode guns required. Works on any smartphone camera via HTML5.
- 100% Offline QR Engine**  
Pure mathematical SVG vector generation works even during complete rural internet outages.
- Official Form 'J' Slip**  
Standardized digital procurement receipts with verifiable QR authentication stamps.

## 4. Presentation & Demo Walkthrough Script

| DEMO STEP            | SCREEN / ACTION  | WHAT TO SAY TO EVALUATORS                                                                                                   |
|----------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Step 1: Introduction | index.html       | "We present KISSAN, a digital solution to solve long Mandi queues and guarantee transparent MSP payouts for farmers."       |
| Step 2: Slot Booking | book-slot.html   | "Farmers reserve arrival windows in advance, receiving an instant, high-resolution vector QR pass and unique token number." |
| Step 3: Gate Scanner | admin-slots.html | "At the Mandi gate, officers scan the QR code with their mobile cameras for immediate identity and vehicle verification."   |
| Step 4: Yard Flow    | admin-queue.html | "The vehicle enters our 3-stage live yard queue, eliminating traffic bottlenecks and providing real-time wait estimation."  |

| DEMO STEP               | SCREEN / ACTION  | WHAT TO SAY TO EVALUATORS                                                                                                         |
|-------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Step 5: Form 'J' Payout | procurement.html | <i>"Upon weighment, an official digital Form 'J' certificate is generated with complete MSP breakdown and DBT bank tracking."</i> |